Credit Scoring Model: LR vs LightGBM, Production Decision

Champion-challenger readout. Operating point: 5% FPR. Offline evaluation on a held-out window.

RECOMMENDATION: SHIP challenger

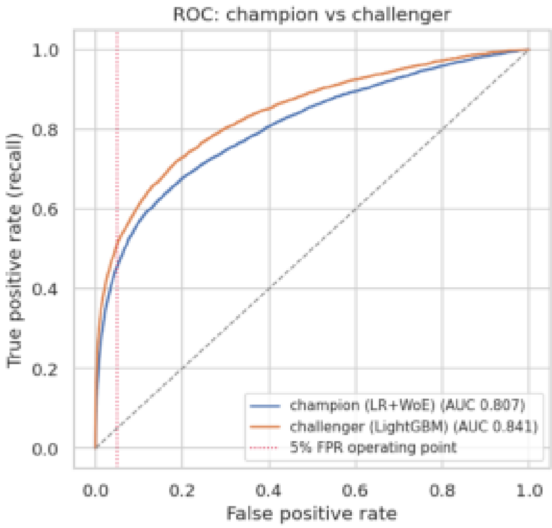
1. Lift at the business threshold

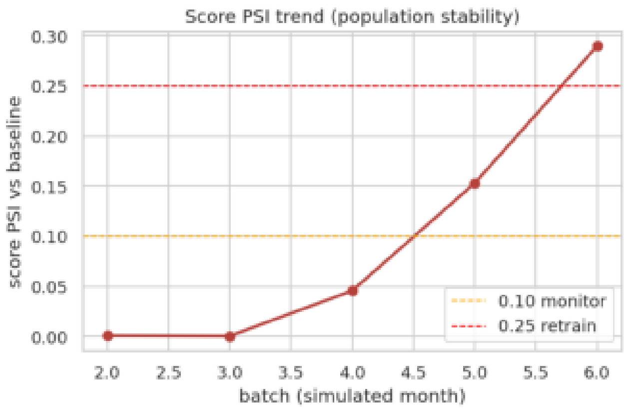
At a 5% FPR operating point, challenger recall 51.1% vs champion 45.5% is +5.6pp more defaulters caught for the same share of good customers declined (95% CI [4.5, 6.8]pp, paired bootstrap, n=1000).

Discrimination: champion AUC 0.807 / KS 0.479, challenger AUC 0.841 / KS 0.533. Rule (pre-registered): ship if lift is at least 3pp and the CI lower bound stays above 0, met.

2. Risk and cost trade-off

Extra defaulters caught (test window): 283, loss avoided $1,415,000 (assumes $5,000/default). False-positive cost about $0, because both models are compared at an identical 5% FPR, so the same share of good customers is declined. Net about $1,415,000 per window.





3. Caveats

- Leakage screen: feature(s) with IV>0.5 = RevolvingUtilizationOfUnsecuredLines; reviewed as the canonical strongest credit driver (revolving utilization), retained, not target leakage.
- Challenger (LightGBM) less interpretable than the WoE scorecard; isotonic calibration applied (see calibration chart).
- Training window is simulated batches; production needs a real vintage.
- Post-deployment monitoring is live: injected-drift sanity check PASSED (CSI caught DebtRatio drift, control age stable).

4. Next step

Ship challenger behind the existing WoE scorecard as a shadow/A-B test; keep champion as fallback. Retrain trigger wired to PSI: investigate at 0.10, retrain at 0.25 (DebtRatio already breached in the drift simulation).

Generated from outputs/results.json, synthetic GMSC-schema data, figures: outputs/roc.png, outputs/psi_trend.png